SOEN 341 Software Process

AI Usage Log

Field	Value
Student	Xavier Lagace (40341466)
Project	CareerConnect : Job Search and Application Tracking Platform
Sprint	Sprint 2
Primary work	Scrum Master / DevOps — full-stack CI and workflow documentation
AI tool	OpenAI coding assistant (Codex through the pi interface)
Log updated	October 26, 2026

1. Purpose of this log

This document records my use of generative AI while reviewing and improving CareerConnect's continuous integration process. AI was used to inspect the repository, explain the existing CI/CD state, propose a full-stack GitHub Actions workflow, draft documentation, and explain how the code change should move through issues, branches, pull requests, and the project board.

I did not accept the output without review. I inspected the generated workflow, compared it with the project's real npm scripts and lockfiles, inspected the intended structure with GitHub's documentation and online Node.js workflow examples, revised the project’s documentation in a separate pull request, and verified the result using local commands and successful GitHub Actions runs.

2. Summary of AI-assisted work

Entry	Task / purpose	Area	Decision
1	Audit the existing CI/CD and development process	Repository analysis	Accepted after verification
2	Expand backend-only automation into full-stack CI	GitHub Actions	Reviewed / Modified
3	Update the README and CI documentation	Documentation	Modified substantially
4	Connect the change to an issue, branch, PR, and project workflow	Scrum / Git workflow	Adapted and applied
5	Validate the final result	Testing / independent verification	Accepted after evidence

3. Implementation record

Work item and pull requests

- GitHub issue: #59 — Add full-stack CI and document the development workflow
- Implementation branch: `chore/task-59-ci`
- Implementation commit: `0047759` — `ci: add full stack checks and document new workflow`
- Implementation pull request: #60, linked with `Closes #59`
- Documentation branch: `chore/update-ci-doc`
- Documentation commit: `f5cdd52` — `chore(docs): update stale readme and add ci doc`
- Documentation pull request: #61

Before and after

Before	After
A workflow named `backend-tests.yml` tested only the server.	`.github/workflows/ci.yml` contains separate backend and frontend jobs.
It ran only for server/workflow changes involving `dev`.	CI runs for pull requests and pushes to `dev` and `main`, and can be started manually.
The frontend test and build scripts were not checked by GitHub Actions.	Vitest and the Vite production build are checked alongside Jest.
CI behavior was not clearly documented.	The README and `docs/CI.md` explain triggers, jobs, commands, and branch flow.

Final CI behavior

The final workflow uses Ubuntu and Node.js 24. Each job checks out the repository, installs dependencies from the appropriate lockfile using `npm ci` , and runs commands from the correct package directory.

- Backend job: `npm test -- --ci` from `server/`
- Frontend job: `npm test -- --run` and `npm run build` from `client/`
- npm caching is configured separately for `server/package-lock.json` and `client/package-lock.json` permissions
- are limited to read-only repository contents
- concurrency cancels an obsolete run when a newer commit is pushed to the same ref

The work implements continuous integration. It does not claim to implement continuous deployment because no production hosting target, deployment credentials, or persistent production storage plan was selected.

4. Detailed AI usage entries

Entry 1 — Audit the existing CI/CD setup

Task ID / title: Issue #59 preparation — repository and process audit
Purpose: Determine whether CareerConnect already had CI or CD and whether user stories/tasks were documented.
AI-suggested content: The AI inspected .github/workflows, both package files, the README, Sprint documentation, Git history, branch state, workflow runs, and deployments. It identified one backend-only GitHub Actions workflow and no deployment workflow.
Validation: I compared the findings with the repository and GitHub interface. The original workflow contained only a server test job, and the GitHub deployments list was empty.
Decision: Accepted after verification.
Reflection: AI accelerated repository discovery, but conclusions about the team's process still had to be checked against the live GitHub repository rather than inferred only from local Markdown files.
Responsible person: Xavier Lagace.

Prompt:
“Inspect the server and client for any CI/CD tests or workflows, indicate the file names if you find any.”

Relevant response/result: The audit found CI for backend tests only, no CD. It identified an original `backend-tests.yml` that I then inspected.

Entry 2 — Create full-stack GitHub Actions CI

Task ID / title: #59 — Add full-stack CI
Purpose: Replace the limited backend workflow with one workflow that verifies both application packages.
AI-suggested content: The AI proposed .github/workflows/ci.yml pull request, push, and workflow_dispatch triggers, read-only permissions, concurrency cancellation, and parallel Backend tests and Frontend checks jobs. It selected the commands by reading the actual scripts in server/package.json and client/package.json.
Validation: I reviewed the YAML line by line and checked its structure against GitHub's official workflow syntax, Node.js build/test guidance, and internet examples. I confirmed that the working directories, package-lock cache paths, Node version, and npm commands match this repository. The AI assistant also ran the same commands locally, and I reviewed the results. Finally, GitHub executed both jobs successfully on the pull request and again after merge.
Decision: Reviewed / Modified. The generated workflow was retained after my review and external crossreference.
Reflection: The important validation was matching it to the repository's real scripts and observing it execute successfully in GitHub Actions. Separating frontend and backend into named jobs also makes a failure easier to locate.
Responsible person: Xavier Lagace.

Prompt evidence:
“Draft a CI test workflow for the front end. Then, consolidate with existing backend CI workflow. Ensure current tests are not touched.”

Relevant AI contribution: Drafted the full-stack YAML file as `ci.yaml`. It was important to understand that it needed to install dependencies on each run because the tests run on VMs hosted by Github.

Entry 3 — Update and revise CI documentation

Task ID / title: CI documentation, PRs #60 and #61
Purpose: Make the README and dedicated CI document describe the workflow accurately without duplicating every API or every project-board item.
AI-suggested content: The AI initially drafted a broader README update and `docs/CI-CD.md` , including the branch flow, local commands, CI triggers, job behavior, branch-protection guidance, and future deployment considerations.

Validation: I reviewed the generated text against the actual workflow and then made my own documentation changes in commit `f5cdd52` I shortened the documentation, removed sections that were broader than needed, and renamed the focused document to `docs/CI.md` .
Decision: Modified substantially.
Reflection: AI produced a complete first draft, but I decided the project needed a shorter CI-focused document. Reviewing links after renaming documentation is part of validating the final result.
Responsible person: Xavier Lagace.

Prompt evidence:
“Draft documentation for the CI workflow. Make sure to include that the existing tests pass and remain valid for authentication, job listing/search, and basic Jobs page states, but automated coverage has not kept pace with every newer feature”

5. Human review and responsibility statement

AI acted as an implementation and explanation assistant. My responsibilities included defining the goal, deciding that the project needed full-stack CI, reviewing the generated YAML, checking external references, confirming repository-specific paths and commands, deciding which documentation to keep, revising that documentation, creating and managing the GitHub issue/branches/PRs, and accepting the result only after the local and hosted checks succeeded.

The final documentation is therefore not a verbatim AI response. Commit `f5cdd52` provides direct evidence of my revisions to the initial draft. The green GitHub Actions runs provide direct evidence that the final workflow executed as intended.

6. References and evidence

1. GitHub Docs, “Building and testing Node.js” — https://docs.github.com/en/actions/use-cases-andexamples/building-and-testing/building-and-testing-nodejs
2. GitHub Docs, “Workflow syntax for GitHub Actions” — https://docs.github.com/en/actions/reference/workflowsand-actions/workflow-syntax
3. CareerConnect workflow — `.github/workflows/ci.yml`
4. CareerConnect CI documentation — `docs/CI.md`
5. CareerConnect README — `README.md`
6. GitHub issue #59 and pull requests #60 and #61
7. GitHub Actions run records linked in Entry 5

7. Final outcome

CareerConnect now has a repeatable full-stack CI workflow that checks backend tests, frontend tests, and the production frontend build on relevant pull requests and pushes. The workflow is documented, has passed repeatedly on GitHub-hosted runners, and was introduced through the same issue/branch/PR process it is intended to protect. Continuous deployment remains outside this task's scope.